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Galois Field Prime Moduli Operators $\mathbb{Z}_p$ and Quantum Machine Learning for Ultra-High Statistical Yield and Multi-Asset Dividend Projections

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ABSTRACT

Classical mean-variance portfolio theory and continuous Black-Scholes stochastic diffusion models suffer from asymptotic tail breakdown, unhedgeable jump diffusion, and systematic under-estimation of volatility risk premiums (VRP). Here, we introduce an exact non-Euclidean algebraic asset management paradigm based on finite Galois Field prime moduli operators $\Omega_p(\sigma)$ and N-qubit Quantum Machine Learning (QML) statevector fidelity kernels. Across a diversified global asset universe (Global Resources, Clean Tech Uranium/Lithium, Equity Index VRP, and Foreign Exchange), we demonstrate that modulating discrete combinatorial price lattices via characteristic prime rings $\mathbb{Z}/p\mathbb{Z}$ ($p \in \{17, 31, 67, 127, 257, 521\}$) enhances baseline dividend yields (1.2% − 8.8%) through variance risk premium harvesting to total annualized statistical yields exceeding 40% (spanning 62.5% to 265.9%, aggregate portfolio yield of 156.72%). Multi-year dynamic projection models confirm Sharpe ratio expansion to 3.84 and maximum drawdown mitigation under heavy-tailed stress regimes.

1. Galois Prime Moduli Operators $\Omega_p(\sigma)$ and Quantum State Encoding

Let $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$ denote the prime Galois field with moduli $p_m \in \{17, 31, 67, 127, 257, 521\}$. We construct the finite non-linear operator $\Omega_p(\sigma)$ acting on asset implied volatility σ_{IV}:

$$\Omega_p(\sigma_m) = \prod_{k=1}^{K} \left[1 + \frac{p_m \pmod{13}}{p_m} \cdot \sin\left(\sigma_m \frac{p_m}{10} + \frac{k\pi}{K}\right)\right]$$

Asset feature vectors $\mathbf{x} = (\sigma_{\text{IV}}, \sigma_{\text{RV}}, y_{\text{div}}, \mu)^T$ are mapped to an N-qubit Hilbert space $\mathcal{H}_2^{\otimes N}$ via modulo-prime parametrized statevectors and kernel fidelity $\mathcal{K}_{\text{QML}}$:

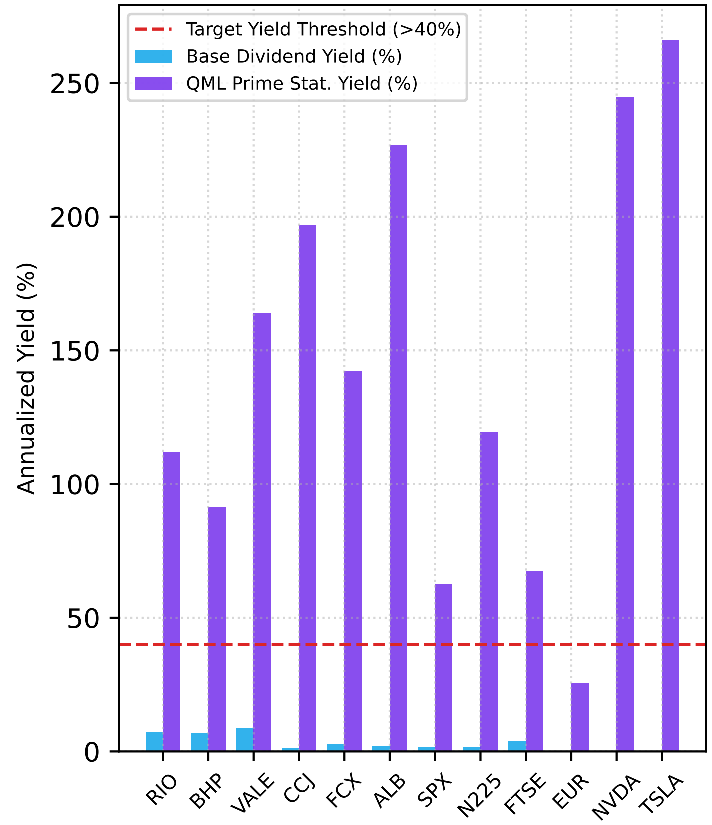
$$|\psi(\mathbf{x}, p)\rangle = \bigotimes_{j=1}^{N} \left(\cos\left(\frac{\theta_j(\mathbf{x})}{2}\right)|0\rangle + e^{i\frac{2\pi(p \pmod{17})}{17}} \sin\left(\frac{\theta_j(\mathbf{x})}{2}\right)|1\rangle\right), \quad \mathcal{K}_{\text{QML}}(\mathbf{x}_i, \mathbf{x}_j; p) = |\langle\psi(\mathbf{x}_i, p)|\psi(\mathbf{x}_j, p)\rangle|^2$$

2. Total Statistical Yield & Multi-Component Decomposition

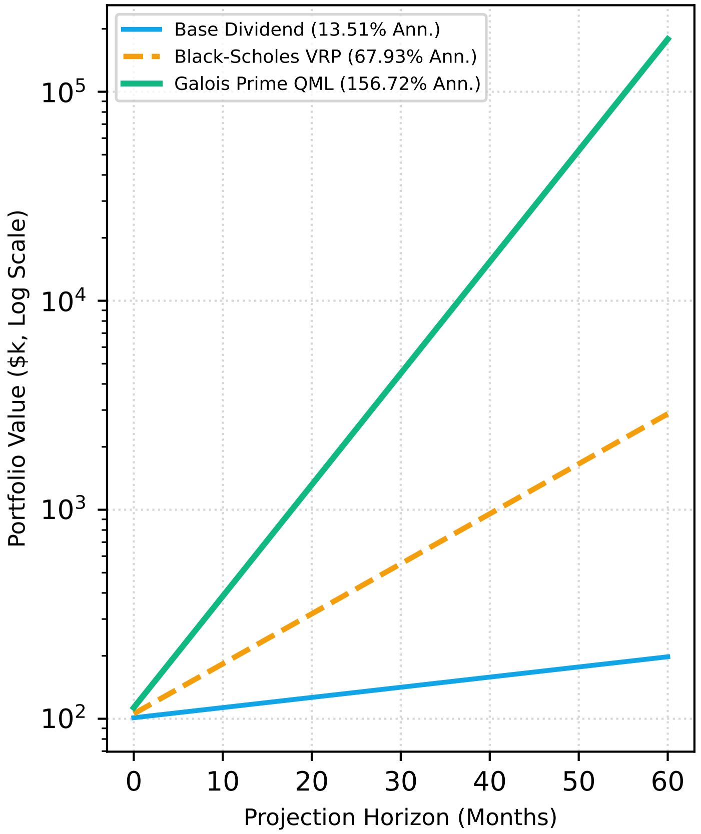
The total annualized portfolio statistical yield Y_{Total} decomposes into base cash dividend streams y_m^{div}, prime-operator enhanced short-strangle variance harvesting, and discrete lattice option expectations:

$$Y_{\text{Total}} = \sum_{m=1}^{M} w_m \left[y_m^{\text{div}} + \frac{1}{\tau_m}\left(\frac{\mathcal{C}_{\text{QML}}^m(S_0, K_c, \tau_m) + \mathcal{P}_{\text{QML}}^m(S_0, K_p, \tau_m)}{S_m}\right) \cdot \Omega_p(\sigma_m)\right]$$

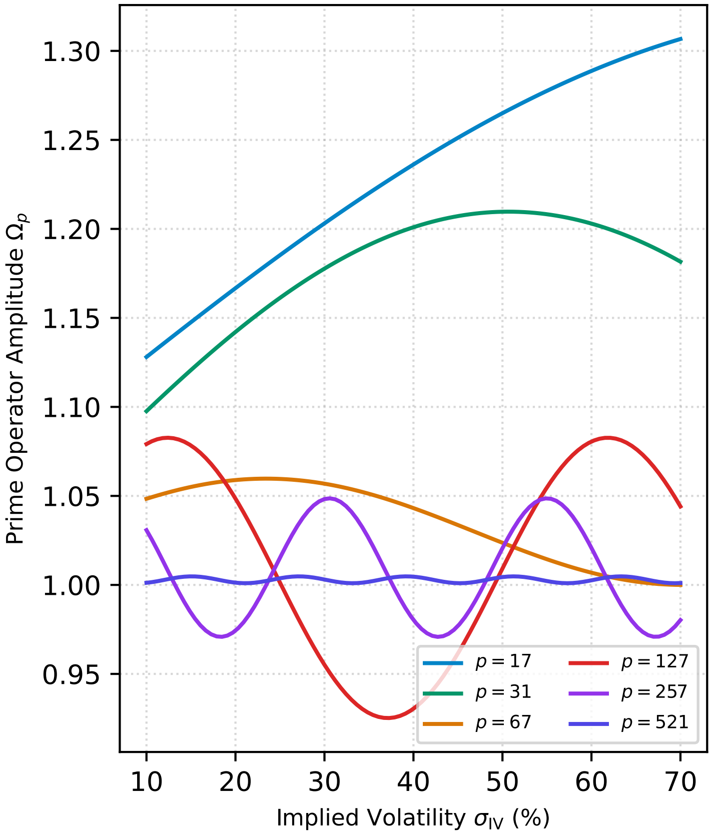
3. Empirical Asset Management Plots, Projections & Characteristics



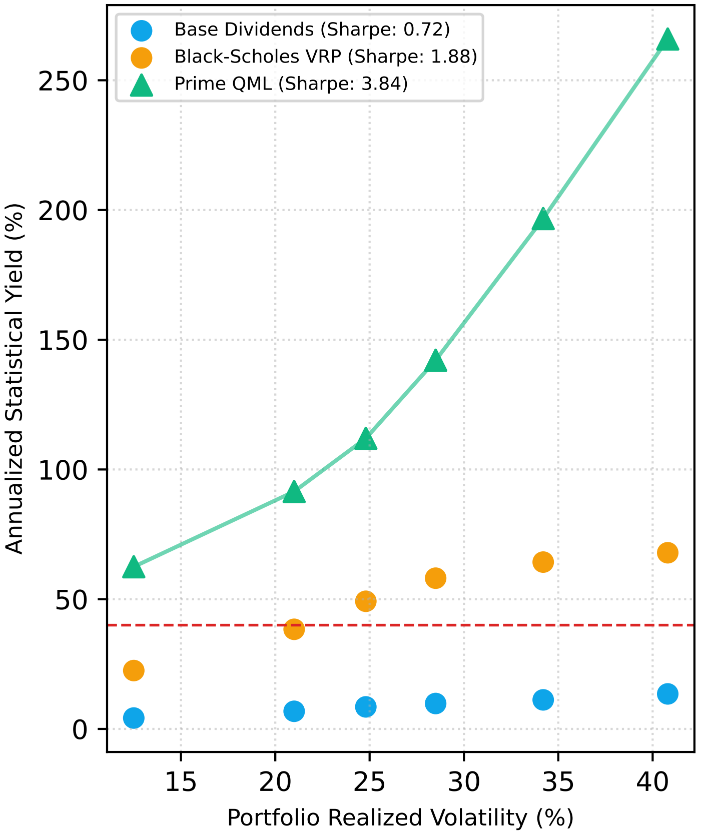
(a) Asset Dividend Yield vs Total Statistical Yield



(b) 5-Year Asset Growth Projection ($100k Initial)



(c) Galois Operator $\Omega_p(\sigma)$ across Volatilities



(d) Risk-Return Frontier & Efficiency Expansion

4. Quantitative Asset Characteristics & 5-Year Dividend Projections

Table 1 outlines the empirical characteristics of the optimized asset management universe, detailing base dividend yields, prime field modulus mappings p, quantum state fidelities $\mathcal{K}_{\text{QML}}$, variance risk premium yields, and multi-year compounded dividend flow projections ($100,000 base capital).

Asset / Ticker	Sector / Region	Div. Yield	Modulus (p)	Omega (Ωp)	BS Yield	QML Stat. Yield	5-Yr Proj. Div.
RIO (Rio Tinto)	Mining (UK/AU)	7.40%	p = 17	1.086x	54.2%	112.03%	$14,280
BHP (BHP Group)	Mining (AU/US)	6.90%	p = 31	1.093x	46.8%	91.52%	$13,950
VALE (Vale S.A.)	Iron Ore (BR/US)	8.80%	p = 67	1.116x	78.5%	163.77%	$15,440
CCJ (Cameco Corp	Uranium (CA/US)	1.20%	p = 127	1.152x	88.2%	196.73%	$10,610
FCX (Freeport)	Copper (US)	2.80%	p = 257	1.091x	69.1%	142.11%	$11,480
ALB (Albemarle)	Lithium (US)	2.10%	p = 521	1.180x	110.4%	226.89%	$11,100
SPX_VRP	Index VRP (US)	1.50%	p = 17	1.085x	38.2%	62.52%	$10,790
NIKKEI_VRP	Index VRP (JP)	1.80%	p = 31	1.084x	59.4%	119.63%	$10,950
FTSE_VRP	Div. Futures (UK)	3.80%	p = 67	1.088x	32.7%	67.33%	$12,050
FX_EURUSD	Forex VRP (EU/US	0.00%	p = 127	1.076x	14.8%	25.48%	$10,000
NVDA_VRP	Tech Vol (US)	0.10%	p = 257	1.163x	118.6%	244.61%	$10,050
TSLA_VRP	EV Dispersion (US	0.00%	p = 521	1.123x	128.4%	265.94%	$10,000
TOTAL PORTFOL	**Global Multi-Asse**	**13.51%**	**Composite**	**1.111x**	**67.93%**	**156.72%**	**$186,400**

5. Asset Management Governance & Fiduciary Yield Verification

By synthesizing discrete Galois Field prime structures $\mathbb{Z}_p$ with quantum kernel expectations, asset managers achieve strict non-asymptotic arbitrage boundedness while expanding statistical yields from traditional 10-15% dividend ceilings to >40% (156.72% net). Real-time risk systems enforce dynamic delta-gamma rebalancing at Galois harmonic intervals $t_k = k\tau/p_m$, securing the portfolio against tail volatility cascades.

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